

Aidan Ryder

✉ aidanryder01@gmail.com ☎ +44 7983048133 📍 London 🔗 aidanryder.uk

ABOUT ME

Design Engineering student at Imperial College London and Kitchener Scholar, with strengths in systems thinking, teamwork and prototyping. I am particularly drawn to complex, open-ended problems where I can research unfamiliar areas, connect ideas across disciplines and develop practical solutions with real-world impact.

EDUCATION

MEng Design Engineering • year 2 average: 68.5% (2:1)

2024 – 2028

Imperial College London

London

- Coursework to date: Human-Centred Design (HCD), Industrial Design (DFMA), UI/UX Design, Mechatronics, Sustainable Design, Systems Thinking, FEA and CFD.

SKILLS

Software

Figma, Fusion 360, Ansys Fluent, Static Structural & Modal, MATLAB, Arduino C++

Design

CAD, Circuit Design, Mechanical Design, Systems Design, Business Modelling, Branding, HCD, DFMA, Soldering

Soft Skills

Teamwork, Communication, Time Management, Presentation Skills

PROJECTS

Axis Pro

- Group: Developed and iteratively refined an electromechanical product for an underserved user group. The final design provided controlled vertical translation and rotational movement for precision painting, alongside packaging, branding and a project video.
- CTO: Led the electromechanical and Design for Manufacture and Assembly (DFMA) aspects of the project, reducing manufacturing time and cost through mechanical simplification and appropriate manufacturing methods. Developed leadership skills through leading meetings, assigning tasks and helping manage the project direction.

Airline Seat System Design

- Group: Dismantled a commercial airline seat, carried out extensive desk research and spoke with industry leaders to understand the seat requirements and wider manufacturing and sales system, with the aim of identifying ways to improve the economic, environmental and social sustainability of the airline seating industry.
- Individual: Proposed a blockchain-based product passport system to improve material traceability and end-of-life recycling while preserving intellectual property and enabling secure data exchange with trusted partners. I also explored improving the speed of disassembly through design choices (such as standardising fasteners) that reduced the number of steps and tools required, making end-of-life disassembly more economically feasible.

Rhythm Rush (A*)

- Group: Gathered user data through 21 street interviews, observations and structured experiments to understand human behaviour in urban spaces and identify opportunities to increase urban play. These insights were used to develop a human-centred concept that promotes interaction between strangers. The result was a body-tracking dance game designed to encourage spontaneous social interaction in event queues.

WORK HISTORY

Work Experience Day, Paddington Robotics

06/2026

- Assembled supermarket robots and discussed design-for-assembly choices, identifying possible improvements. Worked so efficiently that the company cut the planned 4-week assembly programme to 1 week. Learned how asymmetry in design can prevent assembly errors.

London

Crew Member

08/2023 – 07/2024

McDonald's

Wirral

- Worked across multiple roles under high time pressure, often covering 2–3 roles during evening shifts. Trained newer employees to use equipment efficiently and safely.

INTERESTS

Sports

Competed in half marathons, casual climbing and general gym/training.

Reading

Reading history books, classics, philosophy and fiction.

General

Riding motorbikes, playing chess and climbing mountains